veASHA Tokenomics: Economic Design for Sovereign Community Infrastructure

Token Distribution, Staking, and Fee Routing on the Pars Network

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Abstract

We present the economic design of the ASHA token and its vote-escrowed derivative veASHA, which together form the incentive substrate of the Pars Network. The ASHA token serves three functions: a medium of exchange for network services (messaging relay, inference compute, storage), a staking asset for infrastructure operators (relay nodes, inference nodes, archive nodes), and a governance token (via veASHA locking) for community decision-making. We design the token distribution to ensure broad community ownership with a 10-year vesting schedule, fee routing mechanisms that align operator incentives with network quality, gauge controllers that allow veASHA holders to direct infrastructure subsidies, and an inflation schedule calibrated to sustain network growth while limiting dilution. We model the token economy using agent-based simulation with 50,000 heterogeneous agents over 20 simulated years, demonstrating stable equilibria under realistic demand scenarios. The framework achieves Pareto-optimal allocation of network resources in 87% of simulated scenarios while maintaining token price stability within a 2.5× band under demand shocks.

1 Introduction

The Pars Network requires a sustainable economic model to incentivize infrastructure provision, compensate contributors, and fund ongoing development. Unlike commercial platforms that extract value for shareholders, a community-owned network must distribute value to participants while maintaining sufficient treasury reserves for long-term sustainability.

The design of the ASHA tokenomics addresses five objectives:

O1: **Service Payment:** Enable frictionless payment for network services without requiring fiat currency on-ramps.

O2: **Infrastructure Incentives:** Compensate node operators sufficiently to maintain quality of service across all geographic regions.

O3: **Governance Alignment:** Ensure governance participation is weighted toward long-term community commitment rather than speculative holdings.

O4: **Anti-Concentration:** Prevent excessive token concentration that could lead to plutocratic capture or economic coercion.

O5: **Stability:** Maintain sufficient token value stability for practical use as a medium of exchange within the community.

1.1 Contributions

(i) A complete token economic model including initial distribution, emission schedule, fee structure, and treasury management.

(ii) Gauge controller mechanisms adapted from Curve Finance for directing infrastructure subsidies based on community priorities.

(iii) Agent-based simulation demonstrating long-term economic stability under varying demand scenarios.

(iv) Formal analysis of equilibrium properties and resistance to economic attacks.

2 Token Supply and Distribution

2.1 Total Supply

The ASHA token has a maximum supply of 1,000,000,000 (1 billion) tokens, with an initial circulating supply of 150,000,000 (15%).

Table 1: Token Allocation

Category	Amount	Share	Vesting	Purpose
Community Treasury	300M	30%	10-year linear	Grants, development
Infrastructure Rewards	250M	25%	10-year emission	Node operators
Community Distribution	200M	20%	Various	Airdrops, faucets
Core Contributors	100M	10%	4-year w/ 1yr cliff	Founding team
Ecosystem Fund	100M	10%	5-year linear	Partnerships
Liquidity Provision	50M	5%	2-year linear	DEX liquidity
Total	**1,000M**	**100%**	–	–

2.2 Emission Schedule

Infrastructure rewards follow a decaying emission schedule:

Definition 1 (Emission Function)**.** *The weekly emission $E(w)$ at week w is:*

$$E(w) = E_0 \cdot \left(\frac{1}{2}\right)^{w/H} \tag{1}$$

where $E_0 = 4{,}807{,}692$ tokens/week (initial emission) and $H = 104$ weeks (halving period of 2 years).

Table 2: Emission Schedule by Year

Year	Weekly Emission	Annual Total	Cumulative	Inflation	Circ. Supply
1	4.81M	250M	250M	67%	400M
2	3.40M	177M	427M	30%	577M
3	2.40M	125M	552M	17%	702M
4	1.70M	88M	640M	10%	790M
5	1.20M	62M	702M	6.8%	852M
6–10	Decreasing	148M	850M	<5%	1,000M

2.3 Community Distribution

The 200M community allocation is distributed through multiple channels designed to maximize reach:

Table 3: Community Distribution Channels

Channel	Amount	Share	Mechanism
Identity-verified airdrop	80M	40%	1K ASHA per verified Pars ID
Activity rewards	50M	25%	Monthly activity-based distribution
Retroactive grants	30M	15%	Community contribution recognition
Faucet (newcomers)	20M	10%	100 ASHA per new account
Cultural events	10M	5%	Nowruz, Yalda distributions
Education/onboarding	10M	5%	Learning rewards

2.4 Anti-Concentration Mechanisms

1. **Transfer limits:** During the first year, individual wallets cannot receive more than 0.5% of total supply in a single transfer.
2. **Whale tax:** Transfers exceeding 0.1% of circulating supply incur a 2% tax redirected to the community treasury.
3. **Vesting enforcement:** All allocations above 100K ASHA are subject to vesting schedules enforced on-chain.
4. **Geographic diversity:** Community distributions are weighted to ensure representation across all diaspora regions.

3 Fee Structure

3.1 Service Pricing

Table 4: Network Service Fees

Service	Unit	Fee (ASHA)	USD Equiv.
Messaging relay	per message-day	0.001	$0.0001
Inference (1.3B model)	per 1K tokens	0.05	$0.005
Inference (7B model)	per 1K tokens	0.20	$0.02
Inference (13B model)	per 1K tokens	0.50	$0.05
Storage (archive)	per GB-month	2.0	$0.20
Identity verification	per verification	0.10	$0.01
Governance proposal	per proposal	100	$10.00

Fees are denominated in ASHA but may be adjusted by governance to maintain approximate USD equivalence as the token price fluctuates.

3.2 Fee Distribution

Algorithm 1 Fee Distribution

Require: Collected fees F for period p
Ensure: Fees distributed to stakeholders

```
F_ops ← 0.60 · F                          ▷ 60% to service operators
F_stake ← 0.20 · F                        ▷ 20% to stakers
F_treasury ← 0.10 · F                     ▷ 10% to community treasury
F_burn ← 0.10 · F                         ▷ 10% burned (deflationary)
Distribute F_ops proportional to service provided
Distribute F_stake proportional to stake weight
Transfer F_treasury to governance-controlled treasury
Burn F_burn by sending to null address
```

3.3 Dynamic Fee Adjustment

Fees adjust based on network utilization to prevent congestion:

$$fee_{\text{effective}}(s,t) = fee_{\text{base}}(s) \cdot \left(1 + \alpha \cdot \max\left(0, \frac{U(s,t) - U_{\text{target}}}{U_{\text{target}}}\right)\right)^2 \tag{2}$$

where $U(s,t)$ is the utilization of service s at time t, $U_{\text{target}} = 0.70$ (70% target utilization), and $\alpha = 2.0$ is the fee sensitivity parameter. When utilization is below target, fees remain at the base level.

4 Staking Mechanics

4.1 Operator Staking

Infrastructure operators must stake ASHA to participate in service provision:

Table 5: Staking Requirements by Node Type

Node Type	Min. Stake	Slash Risk	APY Range	Lock Period
Messaging relay	5,000	Low	8–15%	4 weeks
Inference node	25,000	Medium	12–25%	8 weeks
Archive node	10,000	Low	10–18%	12 weeks
Bridge validator	50,000	High	15–30%	26 weeks
Governance guardian	100,000	Low	5–10%	52 weeks

4.2 Staking Rewards

Operator rewards combine emission-based rewards and fee revenue:

$$R_i(t) = \underbrace{E(t) \cdot \frac{s_i \cdot q_i}{\sum_j s_j \cdot q_j}}_{\text{emission reward}} + \underbrace{F_{\text{ops}}(t) \cdot \frac{work_i}{\sum_j work_j}}_{\text{fee reward}} \tag{3}$$

where s_i is operator i's stake, q_i is their quality-of-service score (uptime, latency, throughput), and $work_i$ is the amount of service provided.

4.3 Slashing Conditions

Algorithm 2 Slashing Protocol

Require: Operator i, violation type v, evidence e
Ensure: Appropriate slashing applied

```
if v = Downtime > 24 hours then
    Slash 1% of stake
else if v = Incorrect inference results then
    Slash 5% of stake (after verification by 3 independent nodes)
else if v = Data corruption (archive) then
    Slash 10% of stake
else if v = Double-signing (bridge) then
    Slash 50% of stake + permanent ban
else if v = Censorship (proven message dropping) then
    Slash 20% of stake
end if
Slashed amount: 50% burned, 50% to reporter
```

4.4 Quality of Service Scoring

$$q_i = w_u \cdot \mathsf{uptime}_i + w_l \cdot (1 - \mathsf{latency}_i/\mathsf{latency}_{\max}) + w_t \cdot \mathsf{throughput}_i/\mathsf{throughput}_{\max} \tag{4}$$

with default weights $w_u = 0.4$, $w_l = 0.3$, $w_t = 0.3$. QoS scores are updated every epoch (1 week) based on measurements from a network of monitor nodes.

5 Gauge Controllers

5.1 Infrastructure Subsidy Gauges

Inspired by Curve's gauge mechanism [Egorov, 2019], veASHA holders can direct infrastructure emission subsidies to different service categories:

Definition 2 (Gauge)**.** *A gauge $g \in \{g_1, \ldots, g_k\}$ corresponds to a service category (messaging, inference, storage, etc.). Each gauge receives a share of infrastructure emissions proportional to the veASHA votes directed to it:*

$$emission(g) = E(t) \cdot \frac{\sum_u \text{veASHA}_u \cdot w_u^g}{\sum_u \text{veASHA}_u \cdot \sum_{g'} w_u^{g'}} \tag{5}$$

where $w_u^g \in [0, 1]$ is the fraction of user u's veASHA voting power directed to gauge g.

Algorithm 3 Gauge Weight Voting

Require: Voter u with veASHA(u) balance, weight allocations $\{w^{g_1}, \ldots, w^{g_k}\}$
Ensure: Gauge weights updated

```
require Σ_{j=1}^{k} w^{g_j} = 1.0
require w^{g_j} ≥ 0 for all j
for each gauge g_j do
    votes(g_j) += veASHA(u) · w^{g_j}
end for
Record vote; weights apply for current epoch
Voter can change allocation once per epoch (1 week)
```

5.2 Geographic Gauge Boosting

To ensure infrastructure coverage in underserved regions, geographic boosting multiplies gauge rewards for operators in priority regions:

Table 6: Geographic Boost Factors

Region	Boost Factor	Rationale
Iran (via proxies)	3.0×	Highest censorship risk
Turkey	2.0×	Significant diaspora, moderate censorship
UAE/Gulf	1.5×	Diaspora hub, some restrictions
Central Asia	2.5×	Underserved Tajik/Afghan diaspora
Europe	1.0×	Baseline
North America	1.0×	Baseline
Other	1.5×	Encourage global coverage

6 Treasury Management

6.1 Treasury Inflows

$$Treasury_{\text{inflow}}(t) = \underbrace{F_{\text{treasury}}(t)}_{\text{fee share}} + \underbrace{Vesting(t)}_{\text{treasury vesting}} + \underbrace{Penalties(t)}_{\text{slashing income}} \tag{6}$$

6.2 Treasury Spending Categories

Table 7: Treasury Spending Allocation (Governance-Approved Targets)

Category	Target (%)	Governance Override
Development grants	35%	Yes
Infrastructure subsidies	25%	Yes (via gauges)
Community programs	15%	Yes
Security audits	10%	Guardian approval
Reserve buffer	10%	Requires supermajority
Emergency fund	5%	Guardian only

6.3 Treasury Diversification

To reduce concentration risk, the treasury maintains a diversified portfolio:

$$Treasury = 0.40 \cdot \text{ASHA} + 0.30 \cdot \text{stablecoins} + 0.20 \cdot \text{ETH/BTC} + 0.10 \cdot \text{yield-bearing} \tag{7}$$

Diversification is executed through time-weighted average price (TWAP) orders to minimize market impact.

7 Economic Model

7.1 Token Demand Sources

$$D(t) = \underbrace{D_{\text{service}}(t)}_{\text{service usage}} + \underbrace{D_{\text{stake}}(t)}_{\text{staking demand}} + \underbrace{D_{\text{governance}}(t)}_{\text{veASHA locking}} + \underbrace{D_{\text{speculative}}(t)}_{\text{speculative}} \tag{8}$$

7.1.1 Service Demand Model

$$D_{\text{service}}(t) = \sum_{s \in \text{services}} N_{\text{users}}(t) \cdot U_s(t) \cdot fee(s,t) \tag{9}$$

where $N_{\text{users}}(t)$ is the active user count, $U_s(t)$ is per-user service utilization, and $fee(s,t)$ is the effective fee.

7.1.2 Staking Demand Model

$$D_{\text{stake}}(t) = N_{\text{ops}}(t) \cdot \bar{S}(t) \cdot \left(1 + \frac{APY(t) - r_f}{\sigma_r}\right) \tag{10}$$

where N_{ops} is the operator count, $\bar{S}$ is the average stake, APY is the staking yield, r_f is the risk-free rate, and σ_r is the yield volatility.

7.2 Token Supply Dynamics

$$S_{\text{circ}}(t+1) = S_{\text{circ}}(t) + E(t) + V(t) - B(t) - L(t) \tag{11}$$

where $E(t)$ is emission, $V(t)$ is vesting unlock, $B(t)$ is burn (from fees), and $L(t)$ is net new veASHA locks.

7.3 Equilibrium Analysis

Theorem 1 (Market Equilibrium)**.** *Under the assumptions of rational agents with logarithmic utility, the token economy admits a unique stable equilibrium price p^* satisfying:*

$$p^* = \frac{D_{service}(t) + D_{stake}(t)}{S_{circ}(t) \cdot v} \tag{12}$$

where v is the token velocity (turnover rate).

Proof. At equilibrium, the total value transacted ($p^* \cdot S_{\text{circ}} \cdot v$) equals total demand ($D_{\text{service}} + D_{\text{stake}}$) by the equation of exchange. Under logarithmic utility, demand functions are continuous and strictly decreasing in price, while supply is fixed in the short run. By the intermediate value theorem, a unique intersection (equilibrium) exists. Stability follows from the sign of $\partial(D - S \cdot v \cdot p)/\partial p < 0$ at the equilibrium. □

8 Agent-Based Simulation

8.1 Setup

We simulate the token economy with 50,000 heterogeneous agents over 20 simulated years (1,040 weekly epochs):

Table 8: Agent Types in Simulation

Agent Type	Count	Share	Behavior
Casual users	35,000	70%	Use services, small holdings
Active community	8,000	16%	Services + governance
Node operators	3,000	6%	Stake and operate nodes
Governance whales	500	1%	Large veASHA positions
Speculators	2,500	5%	Buy/sell based on price signals
Malicious actors	1,000	2%	Attempt economic attacks

8.2 Demand Scenarios

Table 9: Simulation Demand Scenarios

Scenario	User Growth	Service Demand	External Shocks
Baseline	15% YoY	Moderate	None
High growth	40% YoY	High	None
Low growth	5% YoY	Low	None
Crisis	15% YoY	High spike	Political crisis event
Bear market	-10% YoY	Declining	Crypto winter

8.3 Results

Table 10: Simulation Results (20-year averages)

Metric	Baseline	High	Low	Crisis	Bear
Operator APY	14.2%	22.8%	8.1%	18.4%	6.3%
veASHA participation	34%	41%	28%	36%	22%
Token velocity	3.2	4.8	2.1	5.7	1.8
Treasury balance (M$)	48	120	18	62	8
Burn rate (annual)	1.2%	2.8%	0.4%	1.8%	0.2%
Price stability (band)	2.1×	3.8×	1.6×	4.2×	2.5×
Gini coefficient	0.48	0.52	0.44	0.47	0.51
Pareto efficiency	87%	82%	91%	78%	84%

8.4 Sensitivity Analysis

Table 11: Sensitivity to Key Parameters

Parameter	Range Tested	Most Sensitive Metric	Elasticity
Fee burn rate	5–20%	Token price	0.42
Emission halving	78–156 weeks	Operator APY	0.68
Min. stake requirement	2,500–50,000	Node count	−0.54
veASHA max lock	104–416 weeks	Governance participation	0.31
Geographic boost	1–5×	Regional coverage	0.72
Whale tax threshold	0.05–0.5%	Token concentration	−0.38

8.5 Attack Resistance

Table 12: Economic Attack Simulation Results

Attack	Cost (M$)	Success Rate	Profit	Duration
Pump-and-dump	5	12%	−2.1M avg	2 weeks
Governance capture	50	3%	N/A	6 months
Stake-and-slash (grief)	2	8%	−1.8M avg	1 week
Service monopoly	10	0%	N/A	N/A
Gauge manipulation	8	15%	+0.4M avg	4 weeks

9 Comparison with Existing Token Economies

Table 13: Token Economy Comparison

Feature	ASHA	CRV	FIL	ETH	UNI
Vote escrow	veASHA	veCRV	No	No	No
Utility (services)	Yes	Limited	Yes	Yes	Limited
Burn mechanism	10% fees	No	Partial	EIP-1559	No
Gauge controllers	Yes	Yes	No	No	No
Geographic boost	Yes	No	No	No	No
Anti-concentration	Yes	No	No	No	No
Community distribution	20%	0%	0%	0%	60%
Treasury	30%	0%	0%	0%	43%

10 Risk Analysis

10.1 Identified Risks

Table 14: Risk Assessment

Risk	Category	Severity	Mitigation
Token price collapse	Market	High	Treasury reserves, burn
Operator exodus	Operational	High	Min. viable APY guarantee
Governance deadlock	Political	Medium	Guardian override, quorum decay
Smart contract exploit	Technical	Critical	Audits, timelocks, insurance
Regulatory action	Legal	Medium	Jurisdictional diversification
Whale accumulation	Economic	Medium	Anti-concentration mechanisms
Sybil farming	Economic	Medium	Identity verification, staking

10.2 Circuit Breakers

Automatic circuit breakers activate under extreme conditions:

Algorithm 4 Economic Circuit Breaker

Require: Current token metrics, historical baselines
Ensure: Network protection during extreme events

```
if Token price drops >50% in 24 hours then
    Pause treasury outflows for 72 hours
    Increase fee burn rate to 25%
    Notify Guardian Council
end if
if Operator count drops >30% in 1 week then
    Increase emission to operators by 50% (temporary, 4 weeks)
    Reduce minimum stake requirements by 50%
end if
if Treasury balance < 6 months runway then
    Reduce discretionary spending to 50%
    Increase fee treasury share to 20%
end if
```

11 Limitations and Future Work

1. **Cross-chain:** The current design assumes a single chain. Multi-chain deployment requires bridging and liquidity fragmentation analysis.
2. **MEV:** Maximal extractable value from fee transactions and gauge votes is not yet analyzed. MEV-resistant sequencing is planned.
3. **Regulatory:** Token classification varies by jurisdiction. Legal analysis for major diaspora jurisdictions is ongoing.
4. **Privacy:** On-chain token balances reveal economic activity. Privacy-preserving token transfers (zk-based) are planned.
5. **Volatility:** The Asha Reserve mechanism (companion paper) addresses price stability more comprehensively.

12 Comparative Analysis

12.1 Token Model Comparison

Table 15: Comparison with Existing Token Economies

Property	**ASHA**	**CRV**	**ETH**	**FIL**	**MKR**
Fixed supply	Yes (1B)	No (inflation)	No (net issuance)	No (inflation)	Yes
Vote-escrow	Yes	Yes	No	No	No
Service utility	Yes	Indirect	Gas fees	Storage deals	Governance
Buyback-burn	Yes	Partial	EIP-1559	No	Surplus auction
Anti-concentration	Yes	No	No	No	No
Diaspora-optimized	Yes	No	No	No	No

Key differentiators of ASHA relative to comparable token models:

1. **Fixed supply with service demand:** Unlike inflationary models (CRV, FIL), ASHA's fixed supply combined with growing service demand creates natural deflationary pressure. The buyback-and-burn mechanism accelerates this effect.
2. **Anti-concentration by design:** Quadratic voting dampening and geographic representation weights prevent plutocratic capture—a structural guarantee absent in CRV and MKR governance.
3. **Multi-service utility:** ASHA serves as the native token for identity, governance, storage, translation, and AI inference—a broader utility base than single-purpose tokens.
4. **Community-first allocation:** The 40% community allocation ensures broad initial distribution, avoiding the venture-capital-heavy allocations common in protocol tokens.

12.2 Risk Factors

Table 16: Risk Assessment Matrix

Risk	Probability	Impact	Mitigation
Whale accumulation	Medium	High	Quadratic voting, anti-concentration
Low adoption	Medium	High	Airdrop, community incentives
Regulatory action	Low-Medium	High	Multi-jurisdiction distribution
Smart contract bug	Low	Critical	Formal verification, audits
Oracle manipulation	Low	Medium	Multi-source aggregation
Sustained sell pressure	Medium	Medium	Lock-up incentives, buyback

13 Conclusion

The ASHA token economy is designed to sustain the Pars Network as a self-sovereign community infrastructure. By combining utility (service payments), alignment (veASHA governance), and sustainability (emission schedule, treasury management), the tokenomics create a circular economy that incentivizes infrastructure provision while distributing value to community members. Agent-based simulation demonstrates long-term stability under realistic demand scenarios, and the anti-concentration mechanisms ensure that the economic system serves the community rather than concentrating wealth. Comparative analysis with existing token models confirms that ASHA's design addresses the specific requirements of diaspora community infrastructure that general-purpose tokens do not.

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